Tomato Disease Guide

Simple guide (B1 English): symptoms, causes, treatment and prevention for 10 tomato classes

Class	Name	Type	Severity
0	Tomato Bacterial Spot	Bacteria	High
1	Tomato Early Blight	Fungus	Medium to high
2	Tomato Late Blight	Water mold	Very high
3	Tomato Leaf Mold	Fungus	Medium
4	Tomato Septoria Leaf Spot	Fungus	Medium to high
5	Tomato Spider Mites (Two-spotted spider mite)	Pest	Medium to high
6	Tomato Target Spot	Fungus	Medium
7	Tomato Yellow Leaf Curl Virus (TYLCV)	Virus	Very high
8	Tomato Healthy	Healthy	None
9	Tomato Mosaic Virus (ToMV)	Virus	High

Note: Chemical products and doses are different in each country. Always read the label and ask a local expert (agronomist) before you use any chemical.

0. Tomato Bacterial Spot

Class	0	**Type**	Bacteria
Cause	Bacteria (Xanthomonas)	**Severity**	High

This disease is caused by bacteria. It attacks leaves, stems and fruit. It spreads fast when the weather is warm and wet. Sick fruit cannot be sold.

Symptoms

- Small, dark, wet-looking spots on the leaves. The spots turn brown or black and often have a yellow ring.
- The middle of a spot can dry and fall out, so the leaf looks like it has small holes.
- Rough, brown spots on green fruit.
- Many leaves fall off, and the fruit can get sunburn.

What helps the disease

Warm weather (24-30 C), high humidity, rain and watering from above. The bacteria travel on seeds, young plants, tools and people.

How to fix it

- Remove very sick plants and destroy them. Do not put them in compost.
- Spray a copper product (sometimes mixed with mancozeb) as soon as you see the disease. Repeat every 7-10 days in wet weather. Copper protects the plant but does not cure it.
- Do not work in the field when the leaves are wet.
- Use drip irrigation instead of watering from above.

Prevention

- Buy clean seeds and healthy young plants.
- Do not grow tomatoes in the same place for 2-3 years.
- Choose varieties that resist the disease, if you can.
- Clean your tools and stakes. Remove weeds and wild tomato plants.

1. Tomato Early Blight

Class	1	**Type**	Fungus
Cause	Fungus (Alternaria solani)	**Severity**	Medium to high

This is one of the most common tomato diseases in the world. It usually starts on the old leaves at the bottom and moves up. The plant makes less food and gives less fruit.

Symptoms

- Brown or black spots with rings inside, like a target. There is a yellow area around them.
- The old leaves at the bottom get sick first. They can die and fall.
- Dark, sunken spots with rings on the stem and near the top of the fruit.

What helps the disease

Warm weather (24-29 C), wet and dry days one after another, and weak plants. The fungus lives in the soil and on old plant parts.

How to fix it

- Remove the sick leaves and old plant parts quickly.
- Spray a fungicide (for example chlorothalonil, mancozeb or copper). Follow the label. Change the type of product from time to time.
- Put mulch on the soil so water does not splash dirt onto the leaves.
- Give the plants enough food and water so they stay strong.

Prevention

- Rotate crops. Wait 3 years before you plant tomato, potato, pepper or eggplant in the same place.
- Use stakes and leave space between plants so air can pass.
- Water the base of the plant in the morning, not the leaves.
- Use clean seeds and resistant varieties.

2. Tomato Late Blight

Class	2	**Type**	Water mold
Cause	Water mold (Phytophthora infestans)	**Severity**	Very high

This is a very dangerous disease. In cool and wet weather it can kill all your plants in a few days. You must act fast.

Symptoms

- Large, wet-looking, grey-green spots on the leaves. They turn brown and dry.
- White, soft mold under the leaves when the air is humid.
- Dark brown spots on the stems.
- Hard, brown spots on the fruit. They grow and the fruit rots.

What helps the disease

Cool (10-24 C) and wet weather, fog and dew. The spores travel far in the wind. They also come from sick potatoes.

How to fix it

- Act when you see the first signs. Remove sick plants, put them in a bag and destroy them. Do not compost them.
- Spray a fungicide (for example chlorothalonil, mancozeb or other approved products). Repeat every 5-7 days when the weather stays wet.
- Lower the humidity: remove the bottom leaves and let air move between plants. Do not water from above.
- Pick the healthy fruit early if the disease is close.

Prevention

- Use resistant varieties and healthy young plants.
- Remove wild tomato and potato plants.
- Do not grow tomatoes next to potato fields.
- Check the weather and spray before the disease arrives.

3. Tomato Leaf Mold

Class	3	**Type**	Fungus
Cause	Fungus (Passalora fulva)	**Severity**	Medium

This disease is common in greenhouses and humid places. It damages the leaves. In bad cases you get less fruit.

Symptoms

- Pale green or yellow spots on the top of the leaf.
- Grey-brown or olive-green fuzzy mold under the leaf, below each spot.
- Leaves curl, dry and fall. The plant may make less fruit.

What helps the disease

Very humid air (above 85%), little air movement and mild temperatures (22-25 C). The fungus stays on old plant parts and greenhouse walls.

How to fix it

- Lower the humidity. Open the windows, use fans and keep the greenhouse warm at night.
- Remove the sick leaves and destroy them.
- Spray an approved fungicide (for example chlorothalonil, mancozeb or copper) when you see the first signs.
- Water the base of the plant. Do not wet the leaves.

Prevention

- Plant resistant varieties. Note that new forms of the fungus can beat some of them.
- Leave space between plants and cut extra leaves.
- Clean the greenhouse between crops.
- Rotate crops outside and remove old plants.

4. Tomato Septoria Leaf Spot

Class	4	**Type**	Fungus
Cause	Fungus (Septoria lycopersici)	**Severity**	Medium to high

This is a very common leaf disease. The plant loses many leaves. The fruit is rarely sick, but fewer leaves mean less fruit and more sunburn.

Symptoms

- Many small round spots with a dark brown edge and a grey or light brown center.
- Tiny black dots in the middle of the spots.
- It starts on the bottom leaves, then goes up. The leaves turn yellow and fall.

What helps the disease

Warm (20-25 C), wet and humid weather. Rain splash and watering from above. The fungus lives on old plant parts and wild plants from the same family.

How to fix it

- Remove the sick bottom leaves and old plant parts.
- Spray a fungicide (for example chlorothalonil, mancozeb or copper). Repeat every 7-10 days while the weather is wet.
- Use mulch so soil does not splash on the leaves.

Prevention

- Rotate crops for 2-3 years and remove wild plants from the tomato family.
- Use stakes and leave space between plants.
- Use drip irrigation and water in the morning.
- Use clean seeds and healthy young plants.

5. Tomato Spider Mites (Two-spotted spider mite)

Class	5	**Type**	Pest
Cause	Pest (Tetranychus urticae). It is a tiny animal, not a germ.	**Severity**	Medium to high

Spider mites are very small animals that drink the sap of the plant. They like hot, dry and dusty weather. Their number can grow very fast.

Symptoms

- Tiny yellow or white dots on the top of the leaves.
- Leaves turn bronze or yellow, dry up and fall.
- Fine webs under the leaves and on the stems when there are many mites.
- You can see tiny moving dots with a magnifying glass.

What helps the disease

Hot (above 27 C), dry and dusty weather. Plants that do not get enough water. Strong insecticides can kill the good insects that eat the mites.

How to fix it

- Spray the plants with strong water, especially under the leaves, to wash the mites away.
- Use insecticidal soap or neem oil. Cover the underside of the leaves well. Repeat every 5-7 days.
- If there are very many mites, use an approved miticide. Change the product each time to stop resistance.
- Use good mites that eat the bad ones (for example Phytoseiulus persimilis), mostly in greenhouses.

Prevention

- Water the plants regularly and reduce dust.
- Do not use strong insecticides if you do not need them.
- Remove weeds and check the plants and young plants often.

6. Tomato Target Spot

Class	6	**Type**	Fungus
Cause	Fungus (Corynespora cassiicola)	**Severity**	Medium

This fungus attacks leaves, stems and fruit. It looks like early blight. It is common in warm, humid places and in thick plants.

Symptoms

- Small brown spots with rings and a yellow edge on the leaves. The spots grow and join together.
- Dark, sunken spots on the fruit. The skin can become rough or crack.
- Spots on the stems. Many leaves fall in humid weather.

What helps the disease

Warm (20-30 C), humid weather, wet leaves for a long time and thick leaves. The fungus lives on old plant parts and other plants.

How to fix it

- Remove the sick leaves and old plant parts.
- Spray an approved fungicide (for example chlorothalonil, mancozeb or azoxystrobin). Change the type of product from time to time.
- Cut some leaves and use stakes so air can move. Do not water from above.

Prevention

- Rotate with crops that are not in the tomato family.
- Leave more space between plants.
- Start with clean young plants and keep the field free of weeds.

7. Tomato Yellow Leaf Curl Virus (TYLCV)

Class	7	**Type**	Virus
Cause	Virus. Whiteflies carry it from plant to plant.	**Severity**	Very high

This is one of the worst tomato viruses in warm places. If a young plant gets it, you can lose almost all the fruit. There is no cure.

Symptoms

- Leaves curl upward and have yellow edges.
- The plant stays small and the leaves are small and wrinkled.
- Flowers fall. There is little or no fruit on young sick plants.

What helps the disease

Warm weather and many whiteflies. Whiteflies carry the virus from sick plants and weeds to healthy plants.

How to fix it

- There is no cure. Remove sick plants early. Put them in a bag first so the whiteflies do not escape.
- Control the whiteflies: use approved insecticides, yellow sticky traps and shiny mulch. Change the product from time to time.
- Use insect nets in nurseries and greenhouses.

Prevention

- Plant resistant varieties (look for Ty genes).
- Use virus-free young plants grown under nets.
- Remove weeds and old plants after harvest. Do not grow two tomato crops at the same time next to each other.
- Check for whiteflies from the start.

8. Tomato Healthy

Class	8	**Type**	Healthy
Cause	No disease found	**Severity**	None

This class is a healthy plant. There are no signs of disease. The goal is to keep it healthy with good care.

Symptoms

- Green leaves with no spots, no curling, no mold and no webs.
- Normal growth, flowers and fruit. The fruit is smooth and well shaped.

What helps the disease

Good food, enough water (not too much), fresh air and clean tools keep plants strong against disease.

How to fix it

- You do not need any treatment. Look at your plants at least once a week to find problems early.

Prevention

- Water the base of the plant, in the morning. Do not wet the leaves.
- Leave space between plants, use stakes and cut extra leaves.
- Rotate crops, remove old plant parts and clean your tools.
- Use fertilizer based on a soil test. Use clean seeds and young plants.
- Use mulch and check for pests and early signs of disease.

9. Tomato Mosaic Virus (ToMV)

Class	9	**Type**	Virus
Cause	Virus (Tomato mosaic virus)	**Severity**	High

This virus is very strong and spreads easily by touch. It can stay for years on tools, old plants and seeds.

Symptoms

- Light green and dark green patches on the leaves, like a mosaic.
- Leaves are twisted, curled or narrow. The plant stays small.
- Fruit ripens unevenly, has yellow patches and is smaller. You get less fruit.

What helps the disease

It spreads when people touch plants, with dirty tools, with infected seeds and old plant parts. People who smoke can also spread a similar virus. Signs are worse in cool, dark weather.

How to fix it

- There is no cure. Remove sick plants and destroy them. Do not compost them.
- Wash your hands with soap. Clean tools between plants (for example with 10% bleach).
- Remove weeds and old plant parts that can keep the virus.

Prevention

- Plant resistant varieties (Tm-1, Tm-2 genes) and use virus-free seeds.
- Do not touch plants when they are wet. Work on healthy plants first, then sick ones.
- Clean greenhouse surfaces, stakes and trays between crops. Wash your hands after smoking.
- Rotate crops and remove wild tomato plants.